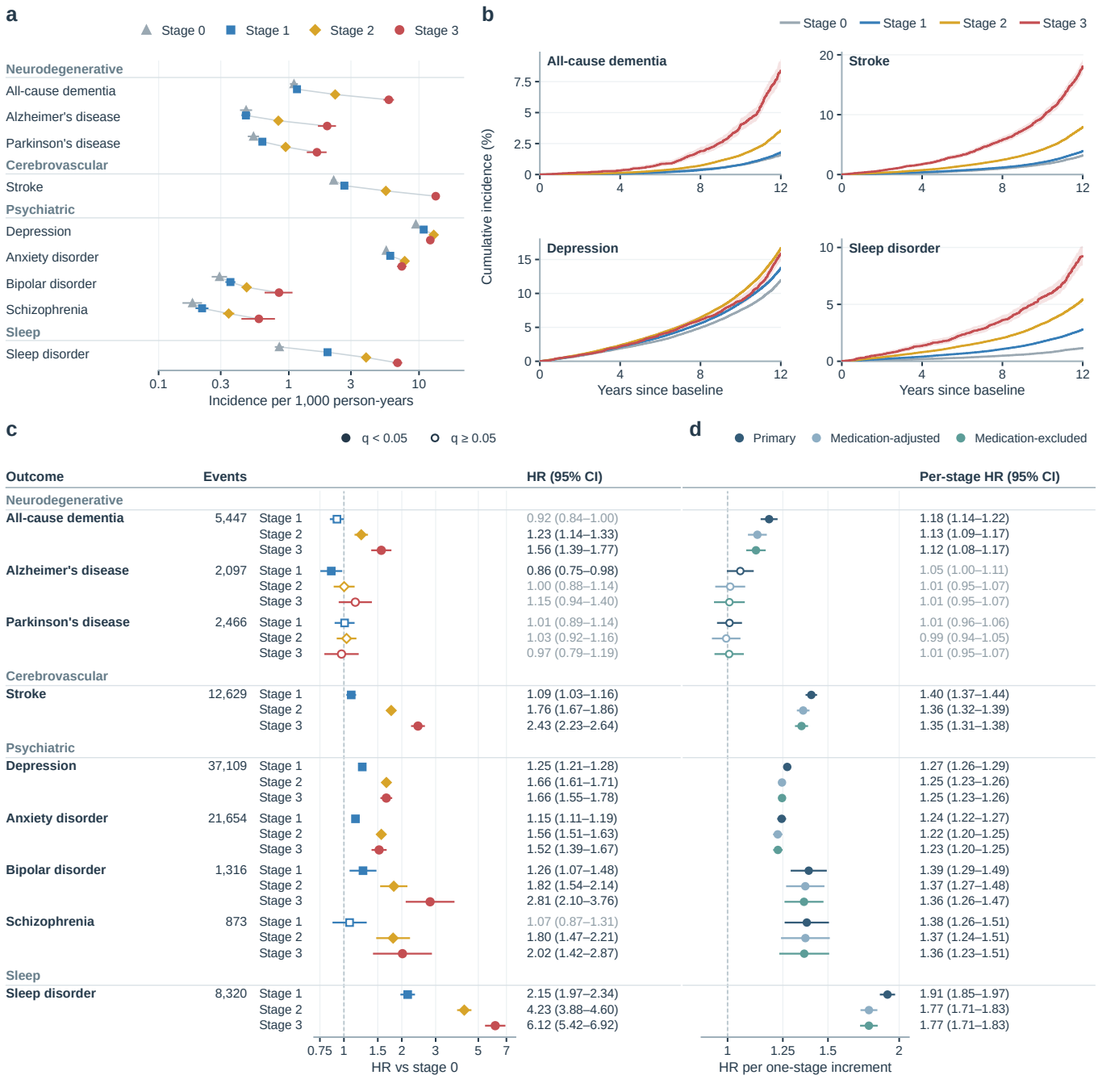

a
Stage 0
Stage 1
Stage 2
Stage 3
Neurodegenerative
All-cause dementia
Alzheimer's disease
Parkinson's disease
Cerebrovascular
Stroke
Psychiatric
Depression
Anxiety disorder
Bipolar disorder
Schizophrenia
Sleep
Sleep disorder
0.1
0.3
1
3
10
Incidence per 1,000 person-years
b
Stage 0
Stage 1
Stage 2
Stage 3
All-cause dementia
Stroke
Depression
Sleep disorder
Cumulative incidence (%)
Years since baseline
Years since baseline
c
q < 0.05
q ≥ 0.05
Outcome
Events
HR (95% CI)
Neurodegenerative
All-cause dementia 5,447
Stage 1 0.92 (0.84–1.00)
Stage 2 1.23 (1.14–1.33)
Stage 3 1.56 (1.39–1.77)
Alzheimer's disease 2,097
Stage 1 0.86 (0.75–0.98)
Stage 2 1.00 (0.88–1.14)
Stage 3 1.15 (0.94–1.40)
Parkinson's disease 2,466
Stage 1 1.01 (0.89–1.14)
Stage 2 1.03 (0.92–1.16)
Stage 3 0.97 (0.79–1.19)
Cerebrovascular
Stroke 12,629
Stage 1 1.09 (1.03–1.16)
Stage 2 1.76 (1.67–1.86)
Stage 3 2.43 (2.23–2.64)
Psychiatric
Depression 37,109
Stage 1 1.25 (1.21–1.28)
Stage 2 1.66 (1.61–1.71)
Stage 3 1.66 (1.55–1.78)
Anxiety disorder 21,654
Stage 1 1.15 (1.11–1.19)
Stage 2 1.56 (1.51–1.63)
Stage 3 1.52 (1.39–1.67)
Bipolar disorder 1,316
Stage 1 1.26 (1.07–1.48)
Stage 2 1.82 (1.54–2.14)
Stage 3 2.81 (2.10–3.76)
Schizophrenia 873
Stage 1 1.07 (0.87–1.31)
Stage 2 1.80 (1.47–2.21)
Stage 3 2.02 (1.42–2.87)
Sleep
Sleep disorder 8,320
Stage 1 2.15 (1.97–2.34)
Stage 2 4.23 (3.88–4.60)
Stage 3 6.12 (5.42–6.92)
HR vs stage 0
d
Primary
Medication-adjusted
Medication-excluded
Per-stage HR (95% CI)
All-cause dementia
1.18 (1.14–1.22)
1.13 (1.09–1.17)
1.12 (1.08–1.17)
Alzheimer's disease
1.05 (1.00–1.11)
1.01 (0.95–1.07)
1.01 (0.95–1.07)
Parkinson's disease
1.01 (0.96–1.06)
0.99 (0.94–1.05)
1.01 (0.95–1.07)
Stroke
1.40 (1.37–1.44)
1.36 (1.32–1.39)
1.35 (1.31–1.38)
Depression
1.27 (1.26–1.29)
1.25 (1.23–1.26)
1.25 (1.23–1.26)
Anxiety disorder
1.24 (1.22–1.27)
1.22 (1.20–1.25)
1.23 (1.20–1.25)
Bipolar disorder
1.39 (1.29–1.49)
1.37 (1.27–1.48)
1.36 (1.26–1.47)
Schizophrenia
1.38 (1.26–1.51)
1.37 (1.24–1.51)
1.36 (1.23–1.51)
Sleep disorder
1.91 (1.85–1.97)
1.77 (1.71–1.83)
1.77 (1.71–1.83)
HR per one-stage increment